CSI 5810 Midterm Project Guidelines

CSI 5810

CSI 5810 Midterm Group Project

Complete Data Science Project Using a Kaggle or UCI Dataset

Assignment overview

The midterm assignment is a group-based, complete data science project. The class will work in seven groups. Each group will select an approved dataset and formulate a clear data-science question that can be answered through data preparation, exploratory analysis, modeling, evaluation, interpretation, and communication.

The project should demonstrate the full data-science workflow rather than only running a machine-learning algorithm. Groups must explain what they did, why they did it, how well the approach worked, what the results mean, and what limitations remain.

Report length: 4–6 pages of main technical content in IEEE conference format. References and a clearly labeled appendix may be placed beyond the main page limit when necessary.

Learning objectives

By completing this project, students should be able to:

- Translate a real-world question into a measurable data-science problem.
- Locate, document, inspect, and responsibly use a public dataset.
- Perform data cleaning, exploratory data analysis, visualization, and feature preparation.
- Establish an appropriate baseline and compare multiple analytical approaches.
- Select evaluation metrics that match the task and the consequences of errors.
- Interpret model behavior and communicate evidence-based conclusions.
- Identify data-quality issues, bias, leakage, reproducibility concerns, and limitations.
- Collaborate effectively in a technical group project.

Group organization

There are seven project groups. Each group should:

- Assign a group coordinator (preferably a graduate student), data-preparation lead, modeling lead, evaluation lead, and report lead. Roles may overlap, but every member must have a **documented technical contribution**.
- Maintain a shared project log containing meeting dates, decisions, task ownership, code contributions, and unresolved issues.
- Use version control or another shared workspace that preserves the history of the work.
- Include a contribution statement in the report describing what each member completed.

What counts as a complete data-science project

Every group must complete the following stages.

1. Problem formulation

Define one focused question rather than attempting to answer several unrelated questions.

Describe:

- The practical or scientific motivation.
- The unit of analysis.
- The input variables and target, outcome, relevance judgment, or discovery objective.
- Whether the task is classification, regression, clustering, association mining, retrieval, dimensionality reduction, or another approved task.
- The intended user or decision supported by the analysis.
- The assumptions and success criteria.

2. Data acquisition and documentation

Document:

- Dataset name and source.
- Public URL and access date.
- Dataset size, attributes, labels, and data types.
- Data-collection context and any known sampling limitations.
- License, privacy, and permitted-use considerations.
- Missing values, duplicates, inconsistent values, outliers, imbalance, and possible leakage.

Do not download or use private, restricted, or personally identifiable data without explicit approval.

3. Exploratory data analysis

Use descriptive statistics and visualizations to understand the data before modeling.

At minimum, include:

- A data dictionary or concise variable summary.
- Target or outcome distribution when a target exists.
- Distributions of important numerical and categorical variables.
- Relationships between important features and the target or between clusters.
- A correlation, similarity, or association analysis when appropriate.
- A written interpretation of the most important patterns and data-quality findings.

Visualizations should have readable labels, units, legends, captions, and a purpose. Do not include charts only because they are easy to generate.

4. Data preparation and feature engineering

Explain every major preparation decision, including:

- Handling of missing values, duplicates, outliers, and invalid records.
- Encoding of categorical values.
- Scaling or normalization of numerical features.
- Text preprocessing when applicable.
- Feature selection, transformation, or construction.
- Train-validation-test splitting or cross-validation.
- Steps taken to prevent information leakage from validation or test data.

All transformations learned from data must be fit only on the appropriate training portion.

5. Baseline and analytical methods

Use at least one simple baseline and at least two meaningful analytical approaches, unless the project is an approved unsupervised or association-mining study with a different comparison design.

Possible baselines include:

- Majority-class prediction.
- Mean or median prediction.
- TF-IDF with cosine similarity.
- A simple k-nearest-neighbors model.
- Linear or logistic regression.
- A basic decision tree.
- A simple clustering or association-rule configuration.

Possible comparison methods include methods covered in CSI 5810, such as k-nearest neighbors, Naïve Bayes, regression, logistic regression, support-vector machines, decision trees, random forests, boosting, clustering, PCA/LSI, association mining, or classical text-mining methods.

For each method, explain:

- Why is it appropriate for the problem.
- Which assumptions it makes.
- Important hyperparameters and how they were selected.
- The expected tradeoff between interpretability, complexity, computation, and performance.

6. Evaluation and comparison

Use evaluation procedures and metrics that match the project type.

Classification may include accuracy, balanced accuracy, precision, recall, F1, specificity, confusion matrices, ROC-AUC, and class-specific performance.

Regression may include mean absolute error, mean squared error, root mean squared error, R-squared, residual diagnostics, and error by subgroup or operating range.

Retrieval may include precision, recall, F1, precision at k, recall at k, mean average precision, ranking comparisons, and qualitative relevance analysis.

Clustering may include silhouette score, within-cluster variation, stability, visualization, external labels when available, and interpretation of cluster profiles.

Association mining may include support, confidence, lift, rule count, redundancy, stability, and usefulness to the stated objective.

Do not report only one score. Explain what the metric measures and why it is appropriate. Report uncertainty, variation across folds, or sensitivity to important choices when feasible.

7. Interpretation, error analysis, and limitations

Explain what the results mean in the context of the original question.

Include:

- Examples of correct and incorrect predictions, relevant and irrelevant results, representative clusters, or useful and weak rules.
- The most important variables, terms, components, or attributes when available.
- Patterns in errors, residuals, class imbalance, subgroup performance, or model instability.
- The effect of changing at least one important modeling choice.

- Limitations involving data quality, sampling, generalization, privacy, fairness, computation, or reproducibility.
- Two or more realistic improvements or next steps.

Required deliverables

1. IEEE-format technical report

Submit one 4–6-page technical report using the IEEE conference paper template and two-column format. The report should include:

- Title, group number, and member names.
- Abstract.
- Index terms or keywords.
- Introduction and problem statement.
- Dataset and data-preparation description.
- Exploratory data analysis.
- Methods and experimental design.
- Results and comparison.
- Error analysis and interpretation.
- Limitations, responsible-use considerations, and conclusion.
- References in IEEE citation style.
- A contribution statement identifying each member's work.

The report should use numbered sections, concise technical writing, figure and table captions, in-text citations, and a reference list. Every figure and table must be discussed in the text. The report should not be a collection of screenshots or uncaptioned output.

2. Reproducible code or notebook

Submit the code or notebook used to produce the analysis and reported results.

The submission should:

- Include setup instructions and required packages.
- Use documented data paths or provide download instructions.
- Set random seeds when randomness affects results.
- Separate data preparation, exploratory analysis, modeling, evaluation, and visualization where practical.
- Generate the principal tables, figures, and metrics used in the report.
- Explain any manual steps or unavailable data that prevent full reproduction.
- Include comments for important decisions rather than relying only on library defaults.

3. Contribution and collaboration record

Submit:

- A contribution statement for every group member.
- A short project log or meeting record.
- A link or archive of the shared code workspace when applicable.
- A statement explaining any generative-AI assistance and for what purpose it was used.

Project selection procedure

There are 21 approved project options organized into seven non-overlapping choice sets of three options each. Each group will receive one choice set and select one project from its three assigned options.

1. Review the three options assigned to your group.
2. Discuss feasibility, data quality, task type, workload, and member interests.
3. Select one final project after considering dataset access, scope, and the group's ability to complete a full data-science workflow and **send me an email with all the group members CC'ed for approval**.

Each dataset should be used by only one group. The project objective should be written as a question that can be evaluated. For example: "Can we predict whether a bank customer will subscribe to a term deposit, and which factors most affect the prediction?" is stronger than "Analyze the Bank Marketing dataset."

Project menu and group assignments

The 21 project choices and the final student-to-group assignments are provided separately on Moodle. Each group has three assigned options and must select one of them.

Minimum analysis expectations by project type

Classification projects

- Use a baseline and at least two comparison models.
- Report a confusion matrix and at least three suitable metrics.
- Discuss class imbalance and the consequences of false positives and false negatives.
- Include an error analysis and, when appropriate, subgroup analysis.

Regression projects

- Use a baseline and at least two comparison models.
- Report at least three suitable error or fit measures.
- Include residual or error analysis.
- Discuss extrapolation, outliers, and the practical meaning of prediction errors.

Clustering projects

- Explain the feature representation and scaling.
- Compare at least two clustering configurations or methods.
- Report internal validation and interpret cluster profiles.
- Discuss stability and whether the clusters are useful for the stated objective.

Retrieval, text-mining, or association-mining projects

- Define relevance, transaction, document, or rule quality clearly.
- Establish a transparent baseline.
- Report task-appropriate ranking, support, confidence, lift, or classification metrics.
- Include qualitative examples and discuss redundancy, noise, or interpretation.

Recommended IEEE report outline

4. Title and author information
5. Abstract
6. Index Terms
7. Introduction
8. Dataset and Problem Definition
9. Data Preparation and Exploratory Analysis
10. Methods and Experimental Design
11. Results and Discussion
12. Error Analysis, Limitations, and Responsible Use
13. Conclusion

14. References

Use the official IEEE conference template. Keep the main report between 4 and 6 pages, use two columns, number sections, cite sources in IEEE style, and ensure that figures and tables are legible at normal viewing size.

Recommended grading rubric

Criterion	Points	Evidence expected
Problem formulation and objective	10	Focused question, motivation, task definition, and success criteria
Dataset documentation and responsibility	10	Source, provenance, data dictionary, quality, privacy, bias, and leakage discussion
Exploratory analysis and visualization	15	Appropriate statistics, meaningful figures, captions, and interpretation
Data preparation and methodology	15	Reproducible preprocessing, baseline, comparison methods, and justified choices
Evaluation and results	20	Valid experimental design, appropriate metrics, comparisons, and uncertainty or sensitivity analysis
Interpretation, error analysis, and limitations	15	Concrete error patterns, model interpretation, limitations, and improvements
Reproducibility and technical report quality	10	Working code/notebook, IEEE format, citations, readable figures, and organization
Group collaboration and individual contribution	5	Project log, contribution statement, and evidence of balanced participation
Total	100	

Academic integrity and generative-AI use

Follow academic-integrity standards and use generative AI responsibly.

- Course notes and approved software libraries may be used, but groups must explain their methods and results.
- Disclose generative-AI assistance used for brainstorming, coding, debugging, editing, analysis, or visualization.
- Verify all generated code, explanations, citations, and results.
- Do not submit work that no group member can explain and defend.
- Include a short disclosure statement in the report describing whether and how generative AI was used.

Final submission checklist

Before submitting, confirm that:

- The project has one focused and measurable objective.
- The group has documented member roles and contributions.
- The dataset source, access date, provenance, license, and limitations are reported.
- EDA includes meaningful statistics and visualizations.
- Data preparation avoids leakage.
- A baseline and at least two meaningful approaches are compared, unless an approved project type uses a different design.
- Evaluation metrics match the task and error costs.
- Results include interpretation, not only scores or screenshots.
- Error analysis includes concrete examples or patterns.
- Limitations, bias, privacy, fairness, and responsible use are addressed.
- The report is 4–6 pages of main content in IEEE conference format.
- References use IEEE style and all external sources are cited.

- Code or a notebook reproduces the principal results.
- The group presentation or narrated walkthrough is included.
- The contribution statement and generative-AI disclosure are included.